Governance Memo

Human-assisted classification of public comments on personalized pricing

From: Ruth & Ketty · INFO 4340 Final Project

Decision: We authorize Prompt B only for assisted aggregate analysis with human review, monitoring, documentation, and escalation. We do not authorize autonomous reporting or final policy judgments.

Purpose and system boundary

We built a generative-artificial-intelligence workflow to classify 1,566 individual public comments into policy-stance and concern categories. The system supports research and triage. It does not determine legal rights, benefits, enforcement outcomes, or the Federal Trade Commission's final policy position. The reconciled human-coded validation set remains the benchmark.

For aggregate research, "unclear" is a valid substantive category and is not automatically forced into a different stance. For case-level interpretation, an unclear record requires human judgment. Records with a model review flag or an evidence quotation that cannot be grounded in the source enter the governance review queue.

Risk matrix

Following the class governance workshop, we scored likelihood from 1 (unlikely) to 5 (expected regularly) and impact from 1 (cosmetic) to 5 (potential harm, legal exposure, or serious analytical distortion). Risk equals likelihood multiplied by impact. These scores are our project-team judgments, not empirical probabilities.

Domain	What could go wrong?	Likelihood	Impact	Risk	Priority
Accuracy and evaluation	Misclassification could distort the public record or a reported policy finding.	4	4	16	High
Human review	Ambiguous or unsupported outputs could bypass analysts.	4	4	16	High
Bias and fairness	Error rates differ across observed text-length groups.	3	4	12	Medium
Accountability	A model output could be treated as a final judgment without a responsible human owner.	3	4	12	Medium
Escalation	A failed threshold could be noticed but not acted on.	3	4	12	Medium
Documentation	Missing prompt/model/output records could make results irreproducible.	2	4	8	Low
Data and privacy	Public comments may still contain personal information sent to an external model service.	2	3	6	Low
Disclosure	Readers may not know where generative artificial intelligence	2	3	6	Low

	was used.				
Security	Secret keys or project files could be mishandled.	1	4	4	Low

Measurable operating rules

The class workshop required rules that could be checked "on a busy Tuesday": each rule needs a metric, threshold, action, and a place where the control lives. We use the following rules.

#	Risk / rule	Required action	Metric	Threshold	Where it lives	Project result
1	Human coding reliability	Revise the codebook and independently recode when Cohen's kappa falls below 0.60.	Cohen's kappa	0.60	Policy + analysis code	Observed 0.706: pass
2	Class-balanced model quality	Do not use automated aggregate reporting when the macro-averaged F1 score is below 0.80.	Macro-averaged F1 score	0.80	Policy + code check	Observed 0.820: pass
3	Substantively important class recall	Do not deploy when recall for a substantively important stance falls below 0.70.	Per-class recall	0.70	Policy + code check	Stronger-action recall 0.921: pass
4	Evidence grounding	Reject, retry, or send to human review when the evidence quotation is not found verbatim in the source comment.	Evidence-quote validity	100% per accepted row	Prompt + code check	29 first-pass failures routed to review
5	Stability	Investigate and retain human review when exact three-run stance agreement falls below 90%.	Three-run exact agreement	90%	Policy + monitoring code	Observed 86.7%: triggered
6	Operational subgroup performance	Investigate when any text-length group's error rate exceeds 1.25 times the best-performing group.	Error-rate ratio	1.25×	Monitoring code	Triggered

Human-review and escalation plan

Row-level review: Send model-flagged records and evidence-grounding failures to a human reviewer. For any case-level interpretation, also review records classified unclear.

Aggregate reporting: Use automated aggregate results only when the performance floor passes and triggered robustness checks are disclosed and monitored.

Threshold failure: Pause expansion, identify the failing subgroup or boundary, review examples, revise the coding rule or prompt only if justified, then revalidate on a held-out or newly coded sample.

Model or prompt change: Treat a material model, prompt, schema, or pricing change as a new system version. Repeat validation and stability checks before relying on the new version.

Audit trail: Retain the prompt text, model identifier, software version, token usage, saved outputs, human gold labels, review flags, and final reporting tables.

Current governance status

0.820	86.7%	510
macro-averaged F1 score - passes 0.80	three-run exact stability - below 90%	combined model/evidence review queue

The review queue contains 510 unique comments (32.6% of the corpus) after combining the model's own review flag with evidence-quotation failures. The 511 comments classified unclear are a different analytical set. We keep unclear as a legitimate aggregate category rather than forcing it into a policy stance; case-level use still requires a human.

Roles and accountability

- We are responsible for the coding dictionary, validation design, reconciliation decisions, interpretation of the results, and final written claims.
- The model may propose classifications and review flags; it does not own or approve a policy conclusion.
- A human analyst must resolve escalated records and document any override or material change to the coding logic.
- Future users must not generalize these results to a different docket, population, model version, or decision context without revalidation.

Why the decision is deliberately conservative

The project demonstrates a useful asymmetry: model inference was extremely inexpensive, but the quality controls were the part that made the output defensible. Prompt B passed our class-balanced performance floor and produced high accuracy on unflagged validation comments, yet the stability and text-length checks still triggered. Those findings are not reasons to hide the model; they are reasons to constrain its role. Our operating decision therefore follows the class principle that governance should live in measurable rules and code checks, not only in a policy paragraph.

Reference points

INFO 4340 Class 17 Governance Checklist Workshop: likelihood × impact risk scoring and threshold-based operating rules.

Federal Trade Commission proposed enforcement policy statement regarding personalized pricing, August 19, 2026.

Project notebook and saved evaluation outputs in the final repository.